

FoodSage: Addressing Recognition Uncertainty in Automated Dietary Monitoring through Human-Robot Dialogue

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Abstract

This paper presents FoodSage, a robotic dietary monitoring system that addresses meal tracking challenges through human-robot dialogue. Unlike conventional approaches reliant on manual logging or fully automated recognition, FoodSage employs a human-in-the-loop methodology combining computer vision with natural conversation. A mobile robot autonomously navigates to dining locations and captures meal images using an onboard camera for automated food recognition. Recognition uncertainty is resolved by conversational verification, facilitated by a large language model (LLM). Through incremental learning, FoodSage incorporates user feedback to improve recognition and adapt to diverse eating scenarios. A pilot study with four participants demonstrates that FoodSage communicates effectively while achieving accurate food tracking. By functioning as both a food monitor and an eating companion, it reduces the burden of manual input, maintains high accuracy, and enriches the dining experience. This work contributes to dining monitoring by offering a human-in-the loop approach, ultimately supporting healthier eating habits.

CCS Concepts

• **Human-centered computing** → **Interaction design**; *Empirical studies in HCI*.

Keywords

Human-in-the-loop, Incremental learning, Diet monitoring, LLM

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1 INTRODUCTION

Maintaining healthy dietary habits has become increasingly challenging in modern society due to fast-paced lifestyles [25]. With the development of sensing technology, dietary monitoring has gained prominence in human-computer interaction (HCI). Smartwatches, necklaces, and other real-time sensing devices are increasingly deployed for dietary tracking and feedback [10, 16, 28, 35].

While dietary monitoring is crucial for supporting healthy eating habits, current methods predominantly rely on manual self-reporting [8, 17, 18, 30]. These methods are time-consuming and susceptible to user forgetfulness, making sustained engagement difficult. Although some researchers have attempted to automate this process through food image recognition, these systems could face technical challenges due to variations in food appearance and presentation styles. Such technological limitations often lead to misclassification, potentially compromising system effectiveness.

To address these limitations, we present FoodSage, a human-in-the-loop robotic dietary monitoring system. Our system consists of a robotic car that automatically navigates to dining locations and tracks food consumption by capturing images through its built-in camera. Acknowledging the inherent uncertainty in food recognition, rather than relying solely on automated processes, FoodSage incorporates interactive human verification. The system initiates natural conversations with users to confirm its observations through human-machine dialogue. Additionally, FoodSage continuously improves its recognition capabilities through incremental learning from user feedback. This approach maintains the convenience of automated tracking while leveraging human knowledge to ensure accuracy.

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We conducted a pilot study with four participants, each completing two seven-minute experimental sessions. After experiencing the system, we collected their user experience with questionnaires and semi-structured interviews. Preliminary results demonstrate that our robot communicates successfully with users and establishes accurate food tracking.

In conclusion, FoodSage provides a dietary monitoring solution that reduces the burden of manual input and addresses recognition uncertainty through human-robot dialogue. It improves tracking efficiency while serving as an eating companion. We hope that this work can serve as a starting point for robotics research in dining, advancing the development of intelligent systems that support healthier eating habits through personalized monitoring and interaction.

2 RELATED WORK

In this section, we present three research areas that have influenced the design of our robots: Human-Robot Interaction in Food, Food Tracking, and Human-in-the-loop.

2.1 Human-Robot Interaction in Food

In the past, many studies have focused on the human-food interaction (HFI). Previous research has explored various design approaches related to food, which can be broadly classified into sensory, emotional, cognitive, and social categories [9, 23, 33]. With the continuous development of hardware devices, human-robot interaction has been applied in dining environments. For example, Khot et al. designed FoBo [13], a robotic companion for solo diners that enhances the experience of solitary dining through celebratory interactions. Jung et al. developed a drinking companion robot that encourages moderate drinking by monitoring the user's drinking pace [12]. Tuyen et al. [31] designed a multimodal humanoid robot capable of helping customers choose food and beverages in a café through interaction. Collectively, these studies demonstrate the potential of dining robots to enrich the dining experience. Meanwhile, large language model (LLM) technology, with its powerful natural language understanding and generation capabilities [5], could significantly improve the interaction experience between humans and robots, particularly in social robot contexts [1, 14]. However, its application in human-food interaction remains relatively limited. To address this gap, our study introduces an LLM-driven robot designed for real-world dining scenarios, aimed at supporting dietary monitoring through human-robot interactions.

2.2 Food Tracking

Early dietary monitoring mainly relied on manual food diaries, where users recorded their daily food intake in detail using paper-based methods [34]. With the advancement of digital technologies, mobile applications have emerged to help users efficiently record their dietary data [8, 17, 18, 30]. For example, the app MyFitnessPal, launched by Under Armour [30], can support users to manually input their dietary data for long-term food tracking. Additionally, some systems have enhanced manual recording by supporting multimodal approaches [4, 21]. For example, Luo et al. presented FoodScrap [21], which combines manual input, photo uploads, and voice

descriptions, offering a more flexible approach to dietary monitoring that improves user experience and data accuracy. However, despite the efficiency of these systems, they still rely on user input, potentially hindering consistent system usage among users.

With advancements in image recognition and sensing technologies, automated dietary monitoring has become a focal point of research [3, 22]. For example, Sen et al. [27] embedded an accelerometer and gyroscope into a smartwatch to detect eating gestures, while simultaneously incorporating camera functionality for food image capture, thereby creating a comprehensive automated dietary monitoring system. However, these approaches could face challenges: environmental factors, such as lighting conditions and background noise, may affect sensor data quality and system reliability. Additionally, the substantial heterogeneity in food appearances and preparation methods presents obstacles for accurate food recognition algorithms.

To balance accuracy and user burden, semi-automated dietary monitoring systems have gained increasing attention in recent years [7, 20]. These systems typically generate food category labels automatically based on captured food photos to record dietary data. In addition, these systems allow users to confirm or correct the generated data. For example, OmniTrack, designed by Kim et al. [15], uses automated triggers to collect dietary data, while allowing users to manually revise food labels. This approach improves data processing efficiency and ensures the accuracy of recognition results. However, these systems often present limitations in user interaction, with a lack of naturalness, which can impact the system's overall effectiveness and long-term user engagement. Drawing from prior research, we developed FoodSage, a semi-automated dietary monitoring system that addresses limitations in existing approaches. Like OmniTrack [15], FoodSage reduces user burden through automation, but extends beyond current semi-automated systems by incorporating natural voice interaction. By incorporating Human-in-the-Loop mechanism with conversational feedback, we aim to provide an intuitive experience while maintaining food tracking accuracy.

2.3 Human-in-the-loop

The human-in-the-loop approach has become an effective mechanism to optimize system performance through real-time user feedback [2, 11, 32]. Many interactive systems leverage the human-in-the-loop mechanism. For example, Sun et al. designed a dialogue agent system for behavior intervention, where the system continuously improves its generative model and the agent itself based on feedback from both the client and the therapist [29]. Similarly, Nakao et al. used user feedback during interactions to address fairness in AI systems [24]. This optimisation process based on user feedback not only enhances the intelligence of the system but also strengthens the interaction between humans and the system. Our designed dietary monitoring robot, FoodSage, was built on these insights, incorporating user feedback into the food recognition dialogue to continuously improve its recognition capabilities.

3 FoodSage

FoodSage is a robotic dietary monitoring system that autonomously recognizes food types and logs dietary data. Unlike fully automated

systems, FoodSage actively engages users in the recognition process to ensure both accuracy and adaptability. When uncertainties arise in food recognition, the system utilizes a combination of large language model (LLM) and text-to-speech (TTS) technology to interact with users, validating results and collecting valuable feedback. This feedback is then used to improve the performance of the system through incremental learning.

FoodSage is based on the realization of a programmable and mobile robot equipped with a camera, on top of which the FoodSage backend is equipped with an image recognition model and a conversation model. The robot autonomously recognizes and locates the position of the food, moves autonomously to the food, takes and transmits an image of the food to the backend, which returns the category of the food and the content of the conversation, and the main robot body carries out the actual conversation with the user. The following will elaborate on the above.

3.1 Robot Hardware

The main body of FoodSage is the TurboPi robot. For food information acquisition, we use the HBVCAM-V2101 camera with 640*480 resolution, which can deliver high-definition images of food to the recognition system. Meanwhile, FoodSage uses wheat wheels and light-emitting ultrasonic sensors to provide the ability to move freely on the table and sense the environment. In order to realize the function of FoodSage voice communication, we also added a sound card. All of the above modules are controlled by a piece of RaspberryPi 4 Model B.

3.2 Car Navigation

TurboPi robot has a basic set of built-in mobility logic, we use its own obstacle avoidance logic as the basis, and independently design the logic for the robot to automatically find its way to the food the user is consuming. The robot first observes the environment on the table, filters out the food closest to the user and considers it to be the food that the user is eating, then moves in that direction, uses ultrasonic detection equipment to control the distance to that food, stops when it reaches 50mm, and calls the camera to take an image of the food.

3.3 Food Recognition

The external computer connects to FoodSage through vncviewer, the food recognition model and the voice interaction model run in the computer, both of which use socket communication. Food recognition is done by our YOLO target detection model trained in advance. We used the ChineseFoodNet dataset for model training [6], this dataset consists of 180,000 high-quality images uploaded by users on major catering platforms, encompassing over 200 typical home-cooked dishes from diverse categories. It captures a wide range of regional dietary variations, reflecting the everyday diets of different regions. The scale, diversity, and regional relevance of the dataset make it suitable for our study. Afterwards, the model was trained for about 30 hours with the support of NVIDIA RTX 4090, the model results after training are as follows: the Precision reaches 0.81, the Recall reaches 0.78, and the mAP50 is 0.86, the overall training results are good. The voice interaction uses GPT o1 with TTS, which realizes the functions of accurate food category

acquisition, diet companion and diet monitoring through the previously mentioned prompt. The incremental learning function uses NVIDIA RTX 4090 to train incremental learning with the acquired data as samples during the robot downtime every night.

3.4 Dialogue Design

To overcome recognition uncertainties, we designed the dialogue module of the system based on GPT o1. The following prompts were utilized.

Role Setting: We set GPT's role as a Chinese food inspection expert to help GPT generate content that better matches the scenario and objectives. GPT will receive the name of the Chinese dish { Food name } and its confidence level { Confidence }. Based on these two pieces of information, it will interact with the user to complete the goal of identifying the food type.

Confidence: Confidence refers to the probability of correctly identifying the dish name. Multi-layered communication strategies are used according to the confidence interval, and this is the core of the prompt design.

Tone Guidance: Tone guidance aims to ensure the generated content strikes a balance between accuracy, warmth, and professionalism. With higher confidence, the response adopts a more affirmative tone, emphasizing accuracy and professionalism. With lower confidence, the response adopts a speculative tone, requiring more friendly content to ask the user about the dish they are eating. Tone guidance is divided into four types based on confidence, as shown in Fig.1.

Intent-Driven: Since the model's primary goal is dietary monitoring, it needs to confirm the food name. However, the user may talk about other topics during their meal. Therefore, the robot should first ensure the user's dining experience. After a proper transition, it should guide the conversation back to the main goal of identifying the dish.

Finally, we fine-tuned the prompt through iterative testing, making slight improvements based on user feedback, particularly in handling special cases, human-machine tone, and phrasing with varying levels of confidence, ensuring the prompt better meets user needs.

3.5 Incremental Learning

Incremental learning involves continuously introducing new data to gradually update and optimize the model, enabling it to adapt to new tasks or data while retaining previously learned knowledge. In FoodSage, the system employs continuous learning, where new feedback data collected through user interaction is constantly integrated with the existing dataset. This mixed dataset is then used to update the food recognition model incrementally. A key challenge during this process is "catastrophic forgetting" [19], where the parameter update mechanism in neural networks can interfere with the weight distribution of previously learned tasks, causing the model to lose proficiency in tasks it had already learned.

To address this issue, we employed a transfer learning approach during the incremental learning process. The system automatically records the food category information provided by the user and adds it as new data to the mixed dataset for the continuous incremental training of the model. By leveraging the knowledge from the old

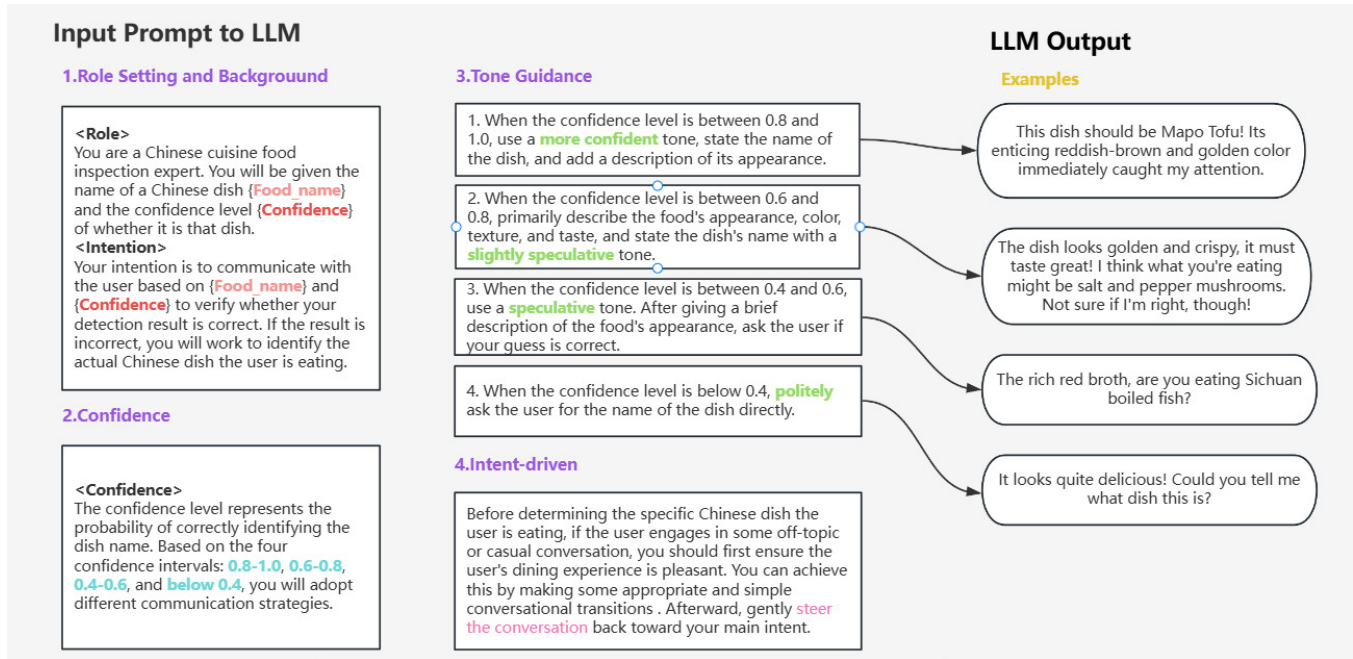


Figure 1: Our designed prompt implementation framework. The diagram outlines the prompt engineering approach with four components: role setting, confidence scoring, tone guidance, and intent management. The framework illustrates how to construct prompts for LLM to identify dishes with varying confidence levels (0.4-1.0), showing corresponding communication strategies and output examples.

dataset as the foundation for the new dataset, we fine-tuned the YOLO model to reduce interference with existing data and enhance the model's ability to recognize new categories. In our training strategy, we employed a method of freezing the lower layers while unfreezing the upper layers. This dynamic learning mechanism enables the system to maintain stable recognition performance when faced with more diverse food scenarios.

4 PILOT STUDY

We conducted a pilot study with four participants (two males and two females, no non-binary or self-descriptions, age: $M=20.29$, $SD=0.70$). None of the participants had prior experience interacting with a robot.

4.1 Procedure

The experiment was conducted in a bright, quiet room free from outside distractions. A dining table was set up in the center of the room and food was prepared in advance for the participants to ensure that they could freely enjoy their meal in a spacious and comfortable dining environment. The FoodSage robotic car was placed on the table and activated when the participants started eating.

Before the experiment began, participants received a detailed explanation of the procedure, which included an overview of the functions of the FoodSage system. During the meal, participants were free to engage in natural conversation with FoodSage while enjoying their food. Each participant took part in two FoodSage

interaction sessions, with each session lasting approximately 7 minutes. To ensure that the pilot study covered a broad range of interactions and provided sufficient data, we selected two types of food, which exhibited significant differences in recognition results within the model. One food item had a high recognition confidence, while the other had a low recognition confidence. After completing both sessions, participants were asked to fill out the User Experience Questionnaire (UEQ) [26], which consists of 26 items, each rated on a scale from 1 to 7. After completing the questionnaire, each participant took part in a semi-structured interview lasting approximately ten minutes. During the interview, participants were asked about their core experiences with FoodSage, including whether the system met their needs, whether they were able to engage in natural conversations, and how they felt about the interaction. All interviews were audio-recorded.

Given the exploratory nature of this pilot study and the limited sample size, the analysis primarily focused on descriptive insights to identify core user experiences to inform system iteration. Quantitative data from questionnaires were analyzed using descriptive statistics without significance testing. Qualitative data from interviews were analyzed inductively, with key user experiences and interaction patterns identified through iterative review.

4.2 Results

Overall, the participants had a positive experience interacting with FoodSage. All participants found the interaction with FoodSage to be enjoyable, and the experimental process was straightforward,

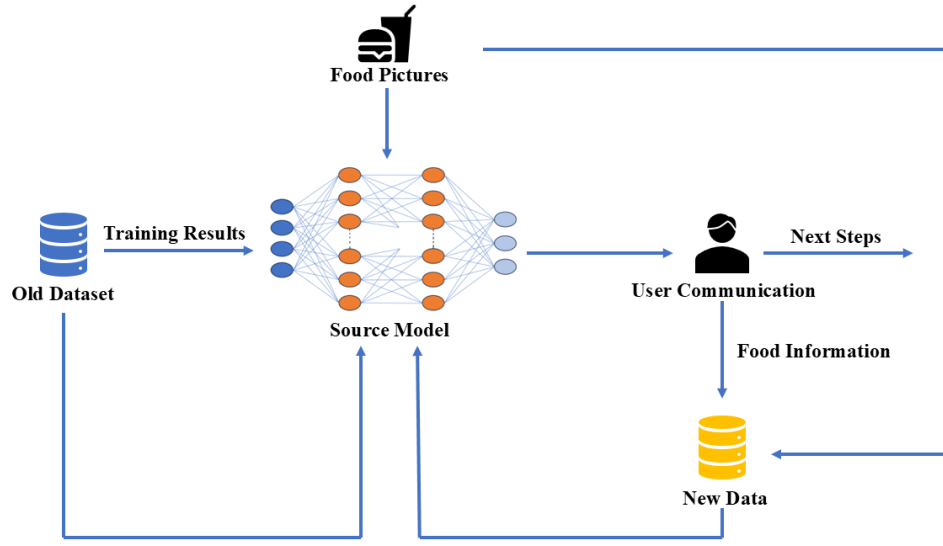


Figure 2: Incremental Learning Process. The model collects user feedback along with relevant food images to form a new data. This new data is then merged with the existing dataset and used for incremental learning to fine-tune the trained model.

with communication being relaxed and easy. The questionnaire results further supported this ($M = 4.73$, $SD = 0.32$). Compared to the benchmark data [13], the scores were relatively high, indicating that participants experienced good pleasure and usability during their interactions with FoodSage.

During the interviews, participants elaborated on their experiences. One participant mentioned: “Its way of speaking was quite cute and the voice was gentle, making the interaction comfortable.” When asked about their opinions on FoodSage’s dietary monitoring function, all participants expressed approval. One participant stated: “I think this feature was great, as it helped monitor my dietary habits, and if I could access the data, it could help improve my eating habits.” Another participant agreed, noting that the feature would be useful for dietary planning.

Regarding the interaction experience, one participant shared: “At first, I simply told it what I was eating, but later we explored different food preparation methods together. Overall, I enjoyed the communication.” Another participant mentioned that FoodSage made them feel cared for during meals. Upon learning that FoodSage could dynamically improve its recognition accuracy based on user feedback, one participant expressed appreciation: “Sometimes, recognition errors occur for various factors, so user corrections are essential.”

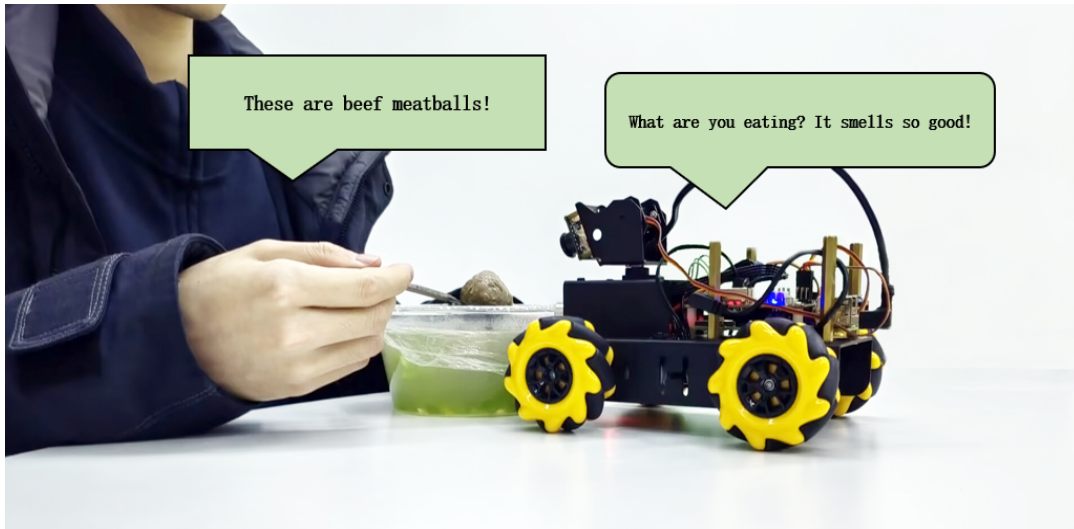
However, participants also provided constructive feedback. Several participants commented that the appearance of FoodSage could be improved. One participant suggested: “I think its appearance could be more appealing and cuter,” while others proposed enhancements such as adding a screen to display animated facial expressions. In terms of communication, three participants noted that the response speed of FoodSage was somewhat slow, and certain interactions felt more like a typical human-computer exchange.

Regarding FoodSage’s recognition capabilities, some participants expressed indifference, but one participant hoped for more recognition categories.

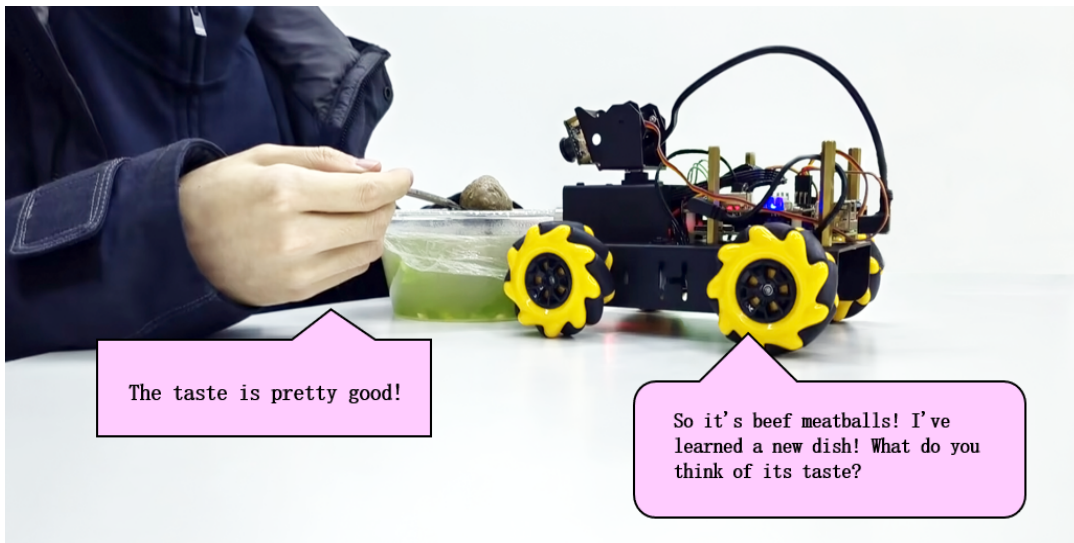
5 DISCUSSION AND FUTURE WORK

The experimental results indicate that our system could provide engaging user experience and efficient dietary monitoring. Despite the strengths of our system, several limitations warrant consideration. First, the robot’s simplistic design may weaken users’ emotional connection. We plan to redesign the robot’s appearance by incorporating cartoon elements and an animated display screen to enhance user engagement. Second, hardware constraints, specifically the low-resolution camera and limited computational power, may result in reduced recognition accuracy and system latency. To address this, we plan to adopt a 1920×1080 or higher resolution camera and improve the system’s computational performance, enabling local execution of the image recognition model to enhance response speed. Additionally, the lack of data visualization tools restricts users from reviewing their dietary records, limiting the system’s potential in health management. We will develop a mobile application synchronized with the backend, allowing users to access and reflect on their dietary data.

Building on these improvements, we plan to conduct a field study with 28-30 participants to gain a more complete understanding of the user experience with FoodSage. By deploying the system in real-world dining scenarios, where factors such as shared utensils or occluded dishes may negatively impact recognition performance, we can better evaluate its practical effectiveness. We will collect both quantitative and qualitative data. Our questionnaire will focus on potential changes in participants’ eating habits throughout the study. Additionally, we will collect participant feedback through



(a) Dialogue 1: In this image, the robot is interacting with the user. After completing the food recognition process, the robot asks the user about the type of food they are eating. The robot says “What are you eating? It smells so good!” and the user responds, “These are beef meatballs!” This dialogue demonstrates how the robot interacts based on its food recognition results and tries to understand the food characteristics through the user’s feedback.



(b) Dialogue 2: In this image, the robot continues the interaction with the user. After the robot identifies the food and learns that the user is eating beef meatballs, it asks “So it’s beef meatballs! I’ve learned a new dish! What do you think of its taste?” and the user responds “The taste is pretty good!” This shows how the robot adjusts its understanding after receiving feedback from the user and continues the interaction to provide a more engaging dining experience.

Figure 3: Human-Robot interaction in a specific scenario: (a) occurs after FoodSage completes the recognition process, where it asks the user about the type of food they are eating based on the confidence level. (b) involves further interaction between FoodSage and the user, aiming to provide a dining companion experience.

semi-structured interviews and employ thematic analysis to extract the most valuable insights. These insights will form the design implications for future intelligent dietary tracking technologies.

6 CONCLUSION

We present FoodSage, a robotic dietary monitoring system designed to address recognition uncertainty through human-robot dialogue and user feedback integration. By leveraging conversational interactions, FoodSage resolves recognition uncertainties and dynamically optimizes its recognition model, enhancing both accuracy and user

trust in AI-driven tracking. Our pilot study demonstrates its feasibility, highlighting its potential to transform dietary monitoring by balancing automation with human insight.

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